

信号作业11 拉氏变换

6-1

(1)

$$e^{at} u(t) \xrightarrow{L} \frac{1}{s-a} \quad \text{Re } s > a$$

极点 a 无零点

(3)

$$e^{-t} u(t) + e^{-2t} u(t) \xrightarrow{L} \frac{1}{s+1} + \frac{1}{s+2}$$

$\text{Re } s > -1$

极点: $-1, -2$

零点: $-\frac{3}{2}$

(5)

$$e^{3t} u(t) + e^{5t} u(t) \xrightarrow{L} \frac{1}{-s+3} + \frac{1}{-s+5} \quad (\text{Re } s < 3)$$

极点: $3, 5$

零点: 4

(3)

$$\sin \pi t u(t) \xrightarrow{L} \frac{\pi}{\pi^2 + s^2} \quad \text{Re } s > 0$$

$$\sin \pi(t-3) u(t-3) \xrightarrow{L} \frac{\pi e^{-3s}}{\pi^2 + s^2} \quad \text{Re } s > 0$$

$$X(s) = \frac{4\pi e^{-3s}}{\pi^2 + s^2} \quad \text{Re } s > 0$$

(8)

$$X(s) = \frac{1000}{100-s^2} \quad -10 < \text{Re } s < 10$$

6-5

$$X(s) = 3 \left(\frac{1}{s} - \frac{1}{s+4} \right) \quad \text{Re } s > 0$$

$$x(t) = 3(1 - e^{-4t}) u(t)$$

$$(3) \quad X(s) = 1 - \frac{3}{s+3} \quad \text{Re } s > 0$$

$$x(t) = \delta(t) - 3e^{-3t} u(t)$$

6-7

5-2

$$(1) \quad u(t) \cdot \sin 2t \xrightarrow{L} \frac{2}{4+s^2} \quad \text{Re } s > 0$$

$$x(t) = \frac{1}{2} \sin 2t u(t)$$

$$-1 \rightarrow -1+2 = -3$$

$$-3 \rightarrow -3+2 = -1$$

收敛域含虚轴

1 与 -1 为极点, 只有双边信号

满足

$x(t)$ 为双边信号

(3)

$$x(t) = \cos 5t u(t)$$

(5)

$$X(s) = \frac{2}{s+3} - \frac{1}{s+2}$$

$$x(t) = 2e^{-3t} u(t) + e^{-2t} u(t)$$

$$Y(s) = \frac{B}{s+1} + \frac{AB}{-s+1}$$

$$= \frac{\frac{1}{2}}{s+1} + \frac{-\frac{1}{2}}{-s+1}$$

$$A = -1 \quad B = \frac{1}{2}$$

3

(1)

$$u(t) \xrightarrow{L} \frac{1}{s}$$

$$u(t-5) \xrightarrow{L} \frac{e^{-5s}}{s}$$

$$X(s) = \frac{1 - e^{-5s}}{s} \quad \text{Re } s > 0$$

6-13

$$H(s) = \frac{2s+3}{s^2+2s+5}$$

$$(1) \quad X(s) = \frac{1}{s} \quad Y(s) = \frac{3}{5s} + \frac{1}{5} \left[\frac{-3(s+1)}{(s+1)^2+4} + \frac{7}{(s+1)^2+4} \right]$$

$$y(t) = \frac{3}{5} \frac{3}{5} u(t) - \frac{3}{5} e^{-t} \cos 2t u(t) + \frac{7}{10} e^{-t} \sin 2t u(t)$$

(2)

$$X(s) = \frac{1}{s+1}$$

$$Y(s) = \frac{1}{4} \frac{1}{s+1} - \frac{1}{4} \frac{s+1}{(s+1)^2+4} + \frac{2}{(s+1)^2+4}$$

$$y(t) = \frac{1}{4} \left(\frac{1}{4} - \frac{1}{4} \cos 2t + \sin 2t \right) e^{-t} u(t)$$

b-17

$$Y(s) = \frac{2}{3} \frac{1}{s-2} + \frac{1}{3} \frac{1}{s+1}$$

$$= \frac{3s}{3(s-2)(s+1)}$$

$$= \frac{s}{(s-2)(s+1)}$$

$$H(s) = \frac{s}{(s-2)(s+1)}$$

$$Re s > -1$$

b-14

$$y''(t) + 4y(t) = x'(t)$$

$$s^2 Y(s) - sy(0) - y'(0) + 4Y(s) = sX(s)$$

$$Y(s) = \frac{sX(s)}{s^2+4} + \frac{sy(0)+y'(0)}{s^2+4}$$

$$X(s) = 0 \Rightarrow Y(s) = \frac{1}{s^2+4}$$

$$y(t) = \frac{1}{2} \sin 2t u(t)$$

$$H(s) = -\frac{1}{s+1} + \frac{2}{s+2}$$

$$y(t) = (-e^{-t} + 2e^{-2t}) u(t)$$

$$Y(s) = \frac{X(s)}{s^2+3s+2} + \frac{y'(0)+(s+3)y(0)}{s^2+3s+2}$$

$$X(s) = \frac{2}{s}$$

$$Y_{zs}(s) = \frac{2}{s(s+1)(s+2)} \quad y_{zs}(t) = (1-2e^{-t}+e^{-2t}) u(t)$$

$$Y_{zi}(s) = \frac{3-5(s+2)}{(s+1)(s+2)} \quad y_{zi}(t) = (e^{-t} + 2e^{-2t}) u(t)$$

$$y(t) = (1 - e^{-t} + 3e^{-2t}) u(t)$$

b-2)

$$H_1(s) = \frac{1}{s} \quad H_1(j\omega) = \frac{1}{j\omega + 1(j\omega)^2}$$

$$H_2(s) = \frac{-W^2}{-W^2 + 2jW + 1}$$

b-15

$$Y(s) = \frac{(s+4)}{(s+1)(s^2+3s+1)}$$

$$y_{zs}(0^+) = \lim_{s \rightarrow \infty} sY(s) = 0$$

$$y_{zs}(\infty) = \lim_{s \rightarrow 0^+} sY(s) = 0$$

b-16

$$\Rightarrow H(s)$$

$$\Rightarrow sH(s) + 2H(s) = \frac{1}{s+4} + \frac{b}{s}$$

$$H(s) = \frac{1}{(s+4)} \left(\frac{1}{s+4} + \frac{b}{s} \right)$$

$$\Rightarrow H(2) = \frac{1}{4} \times \left(\frac{1}{6} + \frac{b}{2} \right) = \frac{1}{6}$$

b=1

$$H(s) = \frac{2}{s(s+4)}$$

b-23

(a)

$$Re s < -2$$

$$-2 < Re s < -1$$

$$Re s > -1$$

因果

稳定

x

x

✓

(b)

$$Re s < -2$$

$$-2 < Re s < -1$$

$$-1 < Re s < 1$$

$$Re s > 1$$

x

x

x

✓

x

6-28 6-30

(1) $H(s) = \frac{ks}{s^2 + (4-k)s + 4}$

(2)

$$s^2 + (4-k)s + 4 = 0$$

$$s_{1,2} = \frac{k-4 \pm \sqrt{k(k-8)}}{2}$$

(2) $H(s) = \frac{s^2}{s^2 + 12s + 32}$

① $0 < k < 8$ 时 $\frac{k-4}{2} < 0$
 $0 < k < 4$

② $k \leq 0$ 或 $k \geq 8$ 时

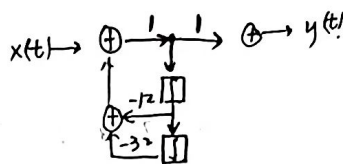
$$\begin{cases} s_1 + s_2 < 0 & k-4 < 0 \\ s_1 s_2 > 0 & k < 4 \end{cases}$$

即 $k \leq 0$

综上 $k < 4$

(3) $k=4$ $H(s) = \frac{4s}{s^2 + 4}$

$h(t) = 4\cos 2t u(t)$



6-30

(1) $H(s) = \frac{-12}{s^2 + 3s + 10}$

